

# Part 1: Capital Bikeshare Demand Analysis & Forecasting

Capital Bikeshare records provide more than a count of bicycle trips: it shows when people travel, how weather changes demand, and how casual and registered riders use the system differently. This analysis uses two years (2011-2012) of hourly Washington, D.C. rental data to answer the complete Part 1 assessment as one connected study. The workflow begins with exploratory analysis, uses those findings to guide statistical modeling, and then develops a forecasting model for the required monthly evaluation windows. The main goal is to move from observed patterns to justified modeling decisions.

## Exploring the Demand Patterns

The dataset contains 17,379 hourly observations. Across the complete two-year hourly timeline, 165 timestamps (about 0.9%) are absent; these were treated as missing observations rather than zero-demand hours. Total hourly demand (cnt) is strongly right-skewed, with a mean of about 189 rentals and a variance of about 32,900. This large difference between the mean and variance immediately suggested that ordinary count assumptions might not be suitable later in the regression analysis. Temperature (temp) and apparent temperature (atemp) were also almost identical statistically ( $r = 0.988$ ), which signaled a possible multicollinearity problem to investigate before modeling.

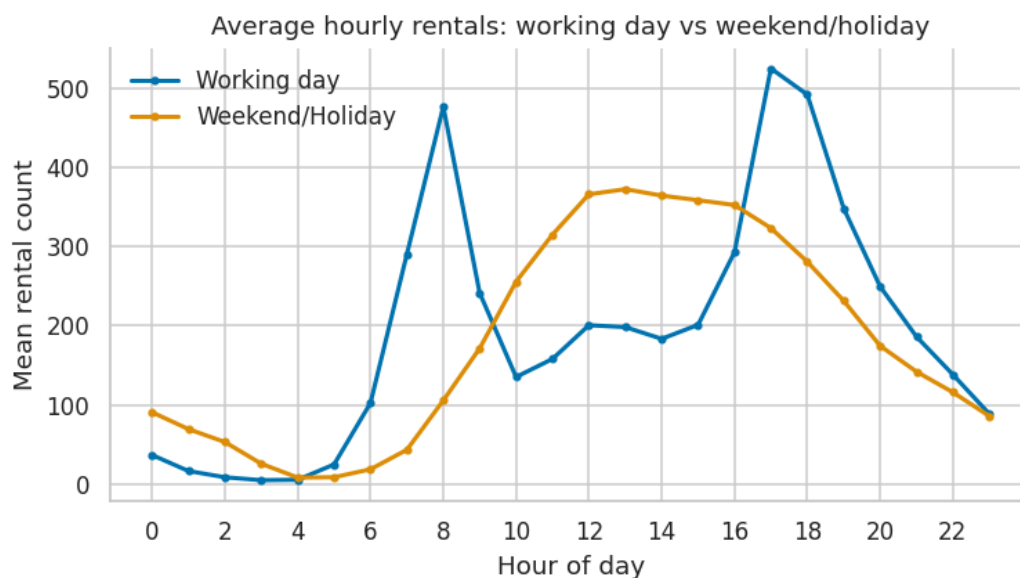


Figure 1. Working days show a bimodal commuter profile (peaks at 08:00 and 17:00-18:00); weekends/holidays show a single broad midday peak.

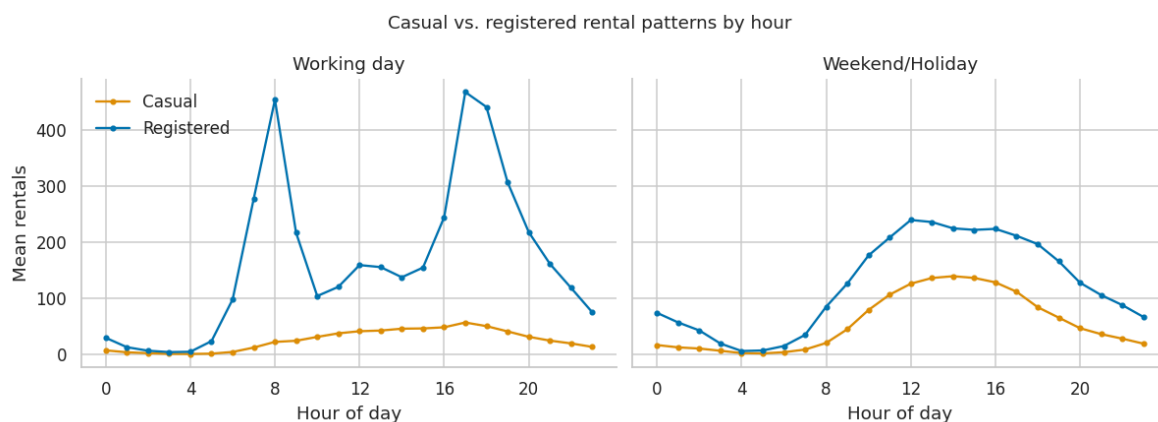


Figure 2. Casual Vs. Registered Rental Patterns By Hour

The exploratory results reveal a clear story about how the system is used. Working days have two sharp peaks, around 8 AM and 5-6 PM, while weekends and holidays have a broader midday peak (Figure 1). Figure 2 shows why: registered riders dominate the commuting peaks, whereas casual riders are more active during leisure hours and on non-working days. Casual demand averages about 57.4 rentals/hour on weekends and holidays compared with 25.6 on working days; registered demand shows the opposite pattern, about 167.6 on working days versus 124.0 on non-working days.

Demand also changes with season and weather. Mean hourly rentals are highest in fall and lowest in spring, and demand generally decreases as weather becomes more severe. The most severe weather category contains only three observations, so it was not used for strong inference. The system also grew substantially: total rentals increased from about 1.24 million in 2011 to 2.05 million in 2012, roughly 65%. One particularly visible anomaly occurs at the end of October 2012, when daily rentals collapse around the Hurricane Sandy period (Figure 3). These patterns motivated the regression model to include time, weather, day type, and year while treating unusual real-world periods as meaningful observations rather than automatically removing them.

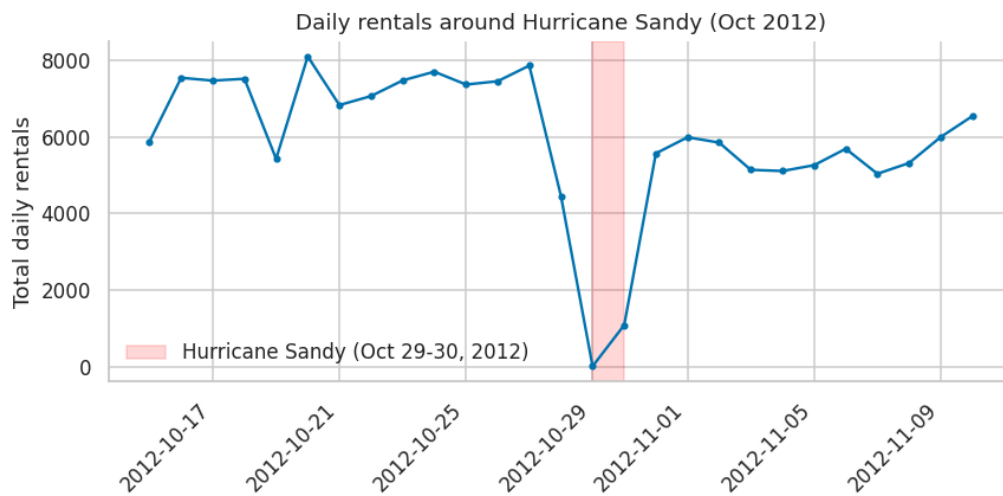


Figure 3. Daily rentals collapse around Hurricane Sandy's landfall (Oct 29-30, 2012).

## From Patterns to Statistical Explanation

The next step was to determine which variables explain rental demand and which statistical model best represents the data. Total hourly rentals (cnt) were modeled using season, weather, temperature, humidity, windspeed, hour, day type, holiday status, and year. Casual and registered demand were also modeled separately to test whether the behavioral differences seen in the figures remained after controlling for other factors.

| Variable  | VIF (with atemp) | VIF (atemp dropped) |
|-----------|------------------|---------------------|
| atemp     | 43.65            | —                   |
| const     | 1.0              | 1.0                 |
| hum       | 1.1              | 1.1                 |
| temp      | 43.55            | 1.01                |
| windspeed | 1.16             | 1.09                |

**Table 1.** Variance Inflation Factors. temp/atemp jointly show VIF  $\approx 43.6$  (severe); dropping atemp reduces VIF to  $\approx 1.0$ -1.1 across all continuous predictors. **atemp is dropped**, keeping temp as the primary thermal predictor.

Table 1 confirms the main multicollinearity concern identified during EDA. temp and atemp have VIF values near 43.6 when used together; after atemp is removed, the remaining continuous predictors have VIF values close to 1.0-1.1. temp was therefore retained and atemp removed. workingday, weekday, and holiday also describe related calendar information, so the primary model used the simpler workingday + holiday representation. A weekday + holiday alternative produced a very similar AIC, supporting the simpler specification.

Model choice was driven by the distribution of the target. OLS was considered first, but about 10% of its predictions were negative, which is impossible for rental counts. Poisson regression avoids negative predictions, but its deviance/df was about 34, far above the value near 1 expected under a suitable Poisson model. The rental data are therefore strongly overdispersed. A Negative Binomial model was selected instead; with  $\alpha = 0.25$ , its Pearson chi-square/df was about 0.98, showing a much better match to the observed count variability. Hour, season, and weather were represented categorically because their relationships with demand are not simple straight-line effects.

| Variable   | IRR   | CI_low | CI_high | p_value |
|------------|-------|--------|---------|---------|
| temp       | 4.364 | 4.067  | 4.682   | <0.001  |
| hum        | 0.823 | 0.78   | 0.869   | <0.001  |
| windspeed  | 0.796 | 0.744  | 0.852   | <0.001  |
| workingday | 0.817 | 0.803  | 0.831   | <0.001  |
| holiday    | 0.742 | 0.708  | 0.779   | <0.001  |
| yr         | 1.604 | 1.579  | 1.629   | <0.001  |

**Table 2.** Negative Binomial GLM, key incidence rate ratios ( $IRR = e^{\text{coefficient}}$ ;  $IRR > 1$  increases expected rentals,  $IRR < 1$  decreases them, holding other variables fixed). Full coefficient table (including all season/weather/hour dummies) is in the accompanying notebook. Model also includes 3 season and 3 weather dummies (IRR range 1.31-1.63 for season; 0.60-0.96 for weather, all except the rare weathersit=4 category significant at  $p < 0.001$ ).

The final Negative Binomial results in Table 2 make the main relationships easier to interpret. Temperature has the strongest positive association: moving from the minimum to maximum normalized temperature is associated with about 4.36 times the expected rentals, holding the other variables fixed. Higher humidity and windspeed are associated with lower expected demand. The year effect is also substantial, with expected demand about 1.60 times higher in 2012 than in 2011, consistent with the growth already visible in the raw data.

The separate models in Table 3 add an important behavioral insight. The working-day IRR is 0.44 for casual riders but 0.93 for registered riders, and temperature sensitivity is much stronger for casual riders. Thus, the regression analysis supports the same story seen in the exploratory plots: casual riders behave more like weather-sensitive leisure users, while registered riders show a more stable commuting pattern.

| Variable       | Casual IRR | Registered IRR |
|----------------|------------|----------------|
| temp (min→max) | 21.68      | 3.27           |
| workingday     | 0.44       | 0.93           |
| holiday        | 0.74       | 0.74           |
| yr             | 1.37       | 1.65           |

**Table 3.** Separate NB GLMs fit to casual and registered rental counts (same predictor set). Working days depresses casual rentals to 44% of their weekend/holiday level, but registered rentals only to 93% — and temp-sensitivity is ~6.6× larger for casual riders. This statistically confirms the behavioral split from Section 1: casual riders are fair-weather, weekend/leisure users; registered riders are comparatively weather- and day-type-insensitive commuters.

Taken together, the exploratory and regression results led to several modeling decisions: atemp was removed because it duplicated temperature information; Negative Binomial regression was preferred over OLS and Poisson because the target is an overdispersed count; hour, season, and weather were treated as categorical effects to capture nonlinearity; and the extremely rare severe-weather category was interpreted cautiously. These decisions were then carried into the forecasting stage where appropriate.

## Forecasting Hourly Bike Demand

For forecasting, the assessment-required split was followed exactly: days 1-20 of every month form the training set and days 21 through the end of the month form the evaluation set. Across all 24 months, this produces 11,460 training hours (66.0%) and 5,919 test hours (34.0%). The central methodological concern was temporal leakage: the model should not receive future rental information that would be unknown at prediction time.

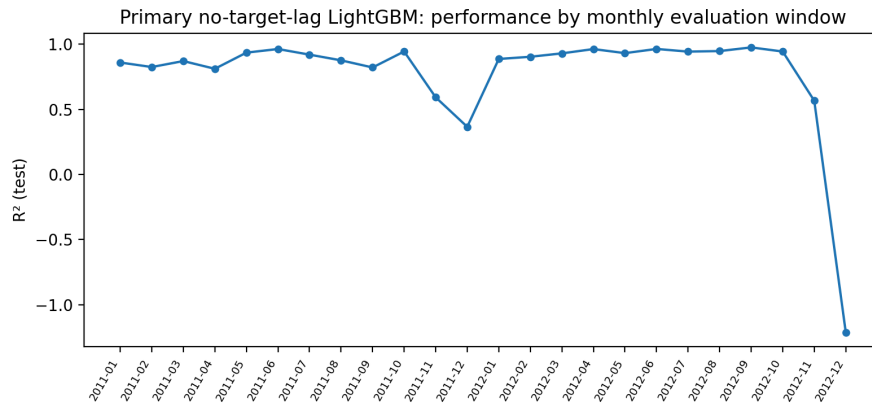
To keep the primary forecast leakage-safe, casual and registered counts were excluded because together they directly reveal cnt. Target-derived lag and rolling-demand features were also excluded from the primary model. The model therefore relied on calendar variables, weather variables, and cyclical encodings of hour, month, and weekday. Since the supplied assessment includes weather variables but no separate weather-forecast dataset, weather was treated as exogenous information available at forecast time, such as from an operational weather forecast.

Performance was evaluated with RMSLE, MAE, RMSE, and  $R^2$  so that both relative and absolute prediction errors could be examined. A Negative Binomial forecast provided an interpretable statistical baseline, while LightGBM was selected as the main machine-learning model because it can learn nonlinear interactions among hour, weather, season, and day type. A lag-based LightGBM and prior-week seasonal-naive model were retained only as rolling-reference analyses because they can use observed demand history from within an evaluation window.

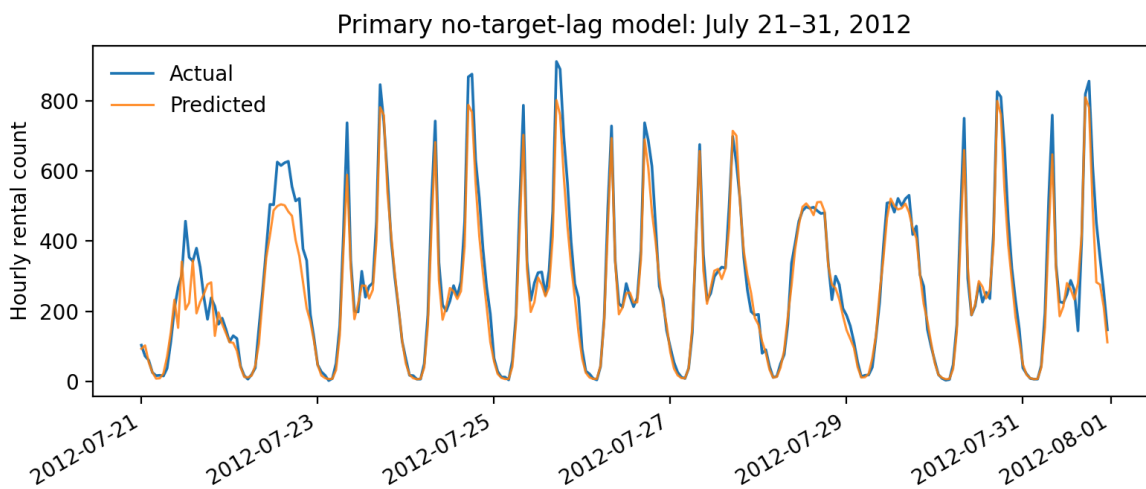
| Model                                    | MAE    | RMSE    | RMSLE | R2    |
|------------------------------------------|--------|---------|-------|-------|
| Seasonal naive (rolling reference)       | 58.16  | 101.672 | 0.651 | 0.685 |
| NB GLM (strict weather+calendar)         | 66.261 | 106.132 | 0.664 | 0.657 |
| LightGBM (supplementary rolling-lag)     | 22.351 | 36.559  | 0.292 | 0.959 |
| LightGBM (PRIMARY: strict no-target-lag) | 33.256 | 56.274  | 0.415 | 0.903 |

**Table 4** summarizes the forecasting results on all 5,919 held-out hours. The primary strict no-target-lag LightGBM achieved MAE = 33.256, RMSE = 56.274, RMSLE = 0.415, and  $R^2 = 0.903$ . This is a strong result because it explains most of the variation in held-out hourly demand without using future target values. The supplementary rolling-lag LightGBM performs better ( $R^2 = 0.959$ ), but it assumes recent observed demand becomes available during the evaluation period and is therefore not used as the main assessment result.

The monthly  $R^2$  plot in Figure 4 shows that performance is not equally strong in every evaluation window, which is important because a single overall score can hide difficult months. Figure 5 gives a more intuitive view: for July 21-31, 2012, predicted demand closely follows the repeated daily structure of the actual rentals, although some sharp hour-to-hour changes remain difficult without autoregressive target history.



**Figure 4.**  $R^2$  of the primary strict no-target-lag LightGBM across the 24 monthly evaluation windows. Performance varies by month, with the weakest windows concentrated around late-year periods; this variability is visible without using actual target values from inside the held-out windows.



**Figure 5.** Actual vs. predicted hourly rentals for July 21-31, 2012 using the primary strict no-target-lag LightGBM. The model captures the recurring daily demand structure from calendar and weather information, while some sharp hour-to-hour changes remain more difficult without autoregressive target history.

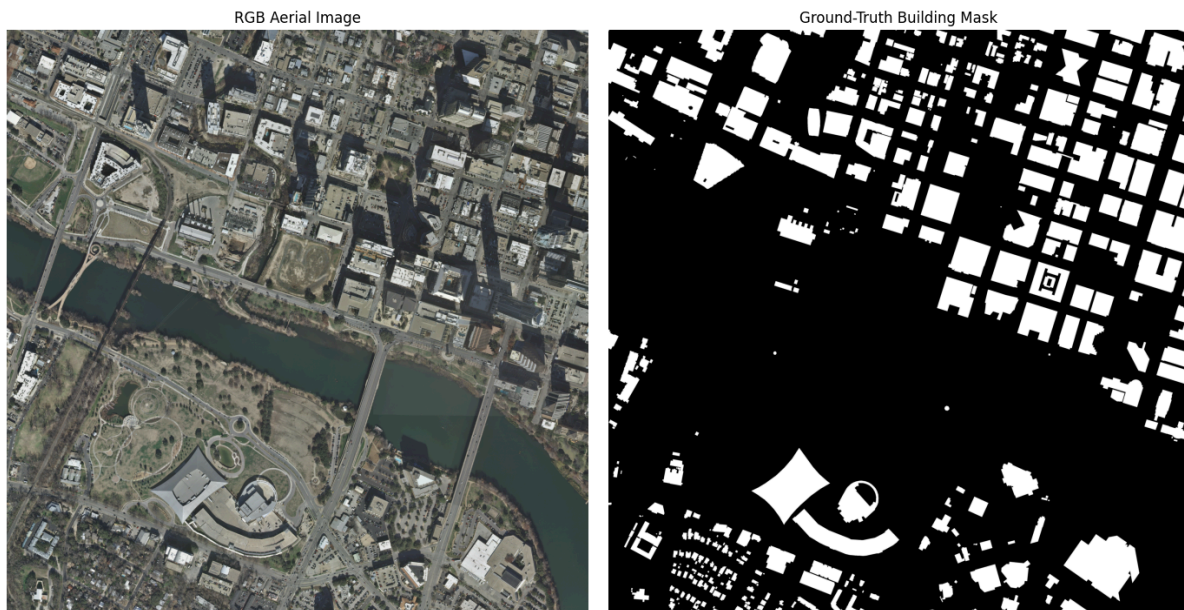
## Part 1 Interpretation and Final Decision

The analysis shows that bike demand is strongly connected to when, why, and under what conditions people use the system. Registered riders show clear commuting peaks around 8 AM and 5–6 PM on working days, while casual riders are more active during daytime and weekends, suggesting more leisure-based use. Weather also influences demand, and the increase from 1.24 million rentals in 2011 to 2.05 million in 2012 shows that the bike-sharing system was growing rapidly. These patterns also helped guide the modeling decisions. Rental counts had much greater variability than expected for a simple count model, making Negative Binomial regression more appropriate for understanding the effects of temperature, humidity, wind, working days, and other factors. The separate analysis of casual and registered riders further confirmed that the two groups respond differently to the same conditions. For forecasting, LightGBM was selected because demand depends on nonlinear relationships among time, weather, season, and day type. The final no-target-lag model achieved  $R^2 = 0.903$  and  $\text{RMSLE} = 0.415$ . A model using recent rental history performed even better, but it was not selected as the primary model because it could use information from within the test period. Therefore, the no-target-lag LightGBM was chosen as the final model because it provided strong forecasting performance while avoiding temporal data leakage.

## Part 2: Building Footprint Segmentation

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The second part addresses a data-analysis challenge of extracting building footprints from high-resolution aerial imagery. The Inria Aerial Image Labeling Dataset contains 180 labeled  $5000 \times 5000$  pixel tiles from Austin, Chicago, Kitsap, Tyrol, and Vienna. Unlike ordinary image classification, the task requires a decision for every pixel, so a useful model must locate buildings accurately while avoiding false building regions and preserving footprint shapes. The analysis was designed as an end-to-end segmentation study, from leakage-safe data partitioning through model selection, training, full-tile prediction, quantitative evaluation, and visual error analysis.



**Figure 6.** A real Inria tile (left) and its ground-truth building mask (right), illustrating the genuine task difficulty: irregular roof shapes/colors, shadows, and dense downtown blocks alongside sparse suburban footprints.

### Methodology and Modeling Decisions

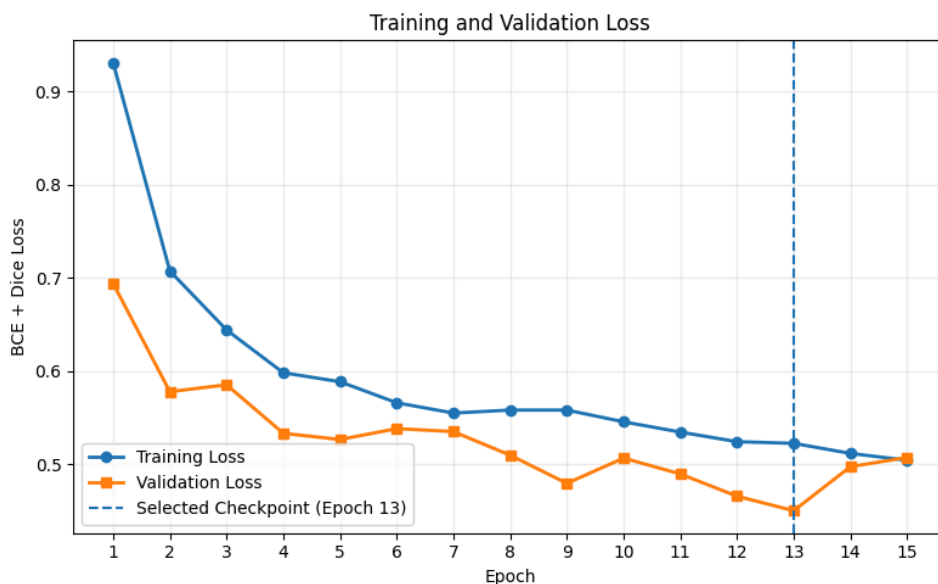
The first decision concerned data partitioning. Randomly splitting  $512 \times 512$  patches could place neighboring pieces of the same large aerial tile in both training and evaluation sets, giving an unrealistically easy test. To prevent this, the split was fixed at the whole-tile level before patch generation. In addition, the entire city of Tyrol (36 tiles) was held out for final testing. The remaining four cities supplied 116 training tiles and 28 validation tiles. This design prevents tile overlap and, more importantly, tests whether the model can transfer to geography it never saw during training.

A U-Net with an ImageNet-pretrained ResNet34 encoder was selected as the segmentation model. U-Net is well suited to this problem because it combines broad image context with detailed spatial information needed to locate building boundaries. The pretrained encoder provides useful low-level visual features such as edges and textures. Training used  $512 \times 512$  RGB patches with ImageNet normalization and random crop, flip, and rotation augmentation so the model encountered varied spatial arrangements during training.

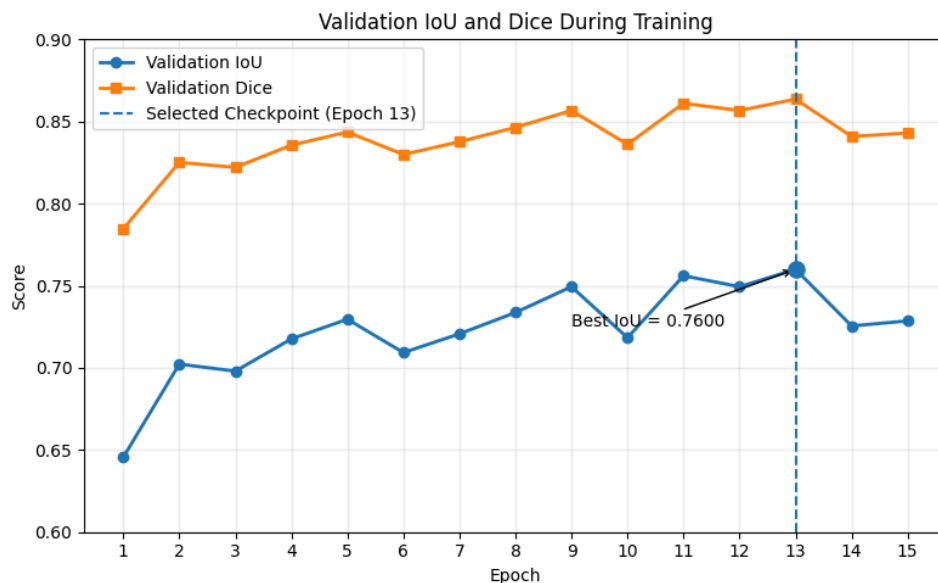
The loss function combined Binary Cross-Entropy (BCE) and Dice loss. BCE provides pixel-by-pixel supervision, while Dice focuses on the overlap between predicted and true building regions and helps when building pixels are much less common than background pixels. The building class occupies about 18.1% of pixels on average. This loss is not

explicitly boundary-aware, so boundary behavior was also examined directly through qualitative predictions and error maps rather than being judged from accuracy alone.

Training continued for up to 15 epochs, with the best checkpoint selected by validation IoU rather than by the final training epoch. The training curves in Figures 7 and 8 show why this matters. Validation IoU reached its best value of 0.760 at Epoch 13, with validation Dice = 0.864. After that point, training loss continued to decrease while validation loss increased, indicating the beginning of overfitting. The Epoch 13 checkpoint was therefore selected for final testing.



**Figure 7.** Training vs. validation loss (BCE + Dice) across 15 epochs. Both generally decrease through epoch 13; validation loss then rises while training loss continues to decline, suggesting the onset of overfitting and supporting selection of the epoch-13 checkpoint.



**Figure 8.** Validation IoU and Dice per epoch. Both metrics climb steadily and plateau around epoch 11-13 with only mild fluctuation thereafter — the checkpoint selection criterion (best validation IoU) lands at epoch 13.

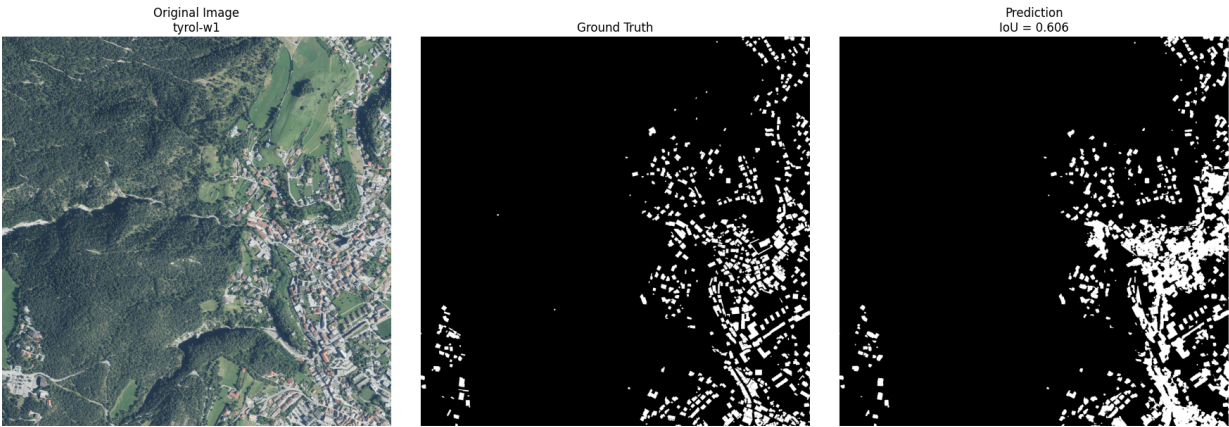
## Results: Performance on the Unseen Tyrol Test Set

Final evaluation was performed on the complete 5000 × 5000 pixel Tyrol tiles rather than only on isolated patches. A sliding-window approach used 512-pixel patches with a 384-pixel stride, averaging probabilities where windows overlapped and reconstructing a full-resolution prediction. Performance was measured with IoU, Dice/F1, Precision, Recall, and pixel Accuracy. IoU and Dice are particularly important here because background pixels dominate the images, meaning accuracy alone can appear high even when building footprints are poorly segmented.

| Metric         | Global (pixel-pooled) | Per-tile mean ± SD |
|----------------|-----------------------|--------------------|
| IoU            | 0.686                 | 0.687 ± 0.064      |
| Dice / F1      | 0.814                 | 0.813 ± 0.045      |
| Precision      | 0.760                 | 0.764 ± 0.086      |
| Recall         | 0.875                 | 0.876 ± 0.028      |
| Pixel Accuracy | 0.977                 | 0.977 ± 0.014      |

**Table 5.** reports performance across all 36 Tyrol tiles, a city never used during training or validation. Global IoU reached 0.686 and Dice/F1 reached 0.814, with Precision = 0.760, Recall = 0.875, and pixel Accuracy = 0.977. The higher recall than precision is informative: the model generally finds real buildings, but it also predicts some extra non-building pixels as buildings. This indicates that over-segmentation and false positives are a more important remaining problem than widespread missed detections.

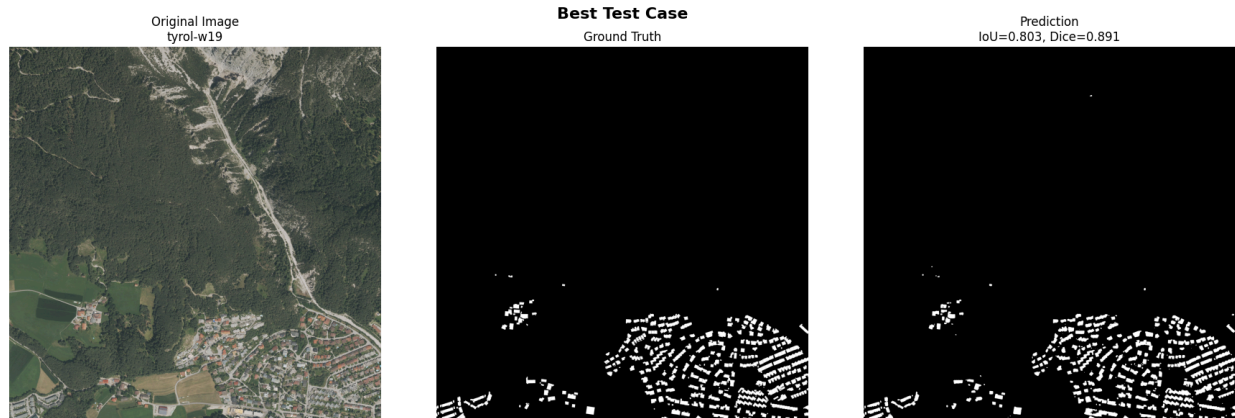
Figure 9 illustrates this behavior on a complete test tile. Although pixel accuracy is 0.960, IoU is only 0.606. The difference demonstrates why accuracy alone is not sufficient for this task: a model can classify the large background area correctly while still making meaningful building-shape errors. The prediction captures most true buildings (Recall = 0.892) but connects or expands some building regions, reducing precision.



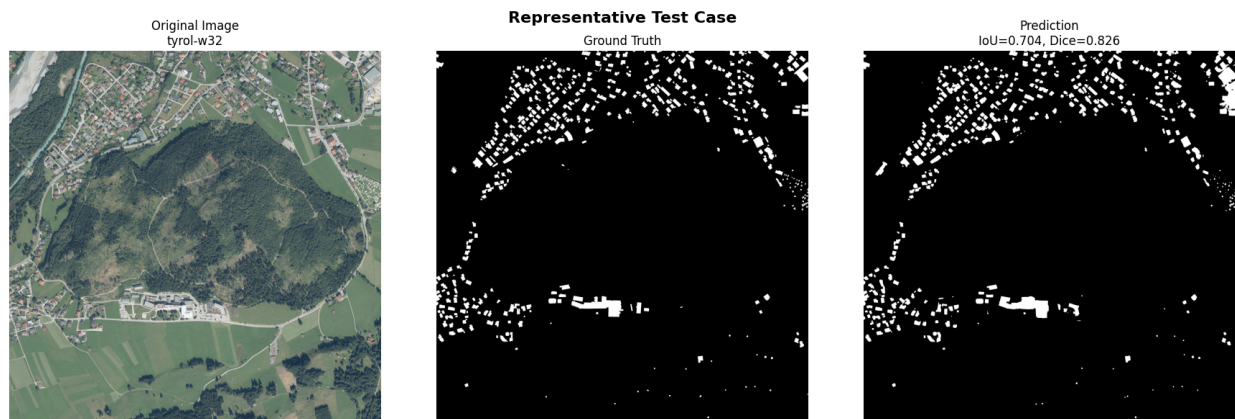
**Figure 9.** Single-tile deep dive (tyrol-w1, stitched full-resolution prediction): IoU=0.606, Dice=0.754, Precision=0.654, Recall=0.892, Accuracy=0.960. The model correctly flags nearly all real buildings (high recall) but over-segments the dense settlement on the right, connecting adjacent roofs into larger predicted blobs (lower precision) — this is the dominant error mode. Pixel accuracy (96.0%) looks excellent despite the modest 0.606 IoU, the exact accuracy-is-misleading effect the multi-metric evaluation above is designed to surface.

## Visual Error Analysis

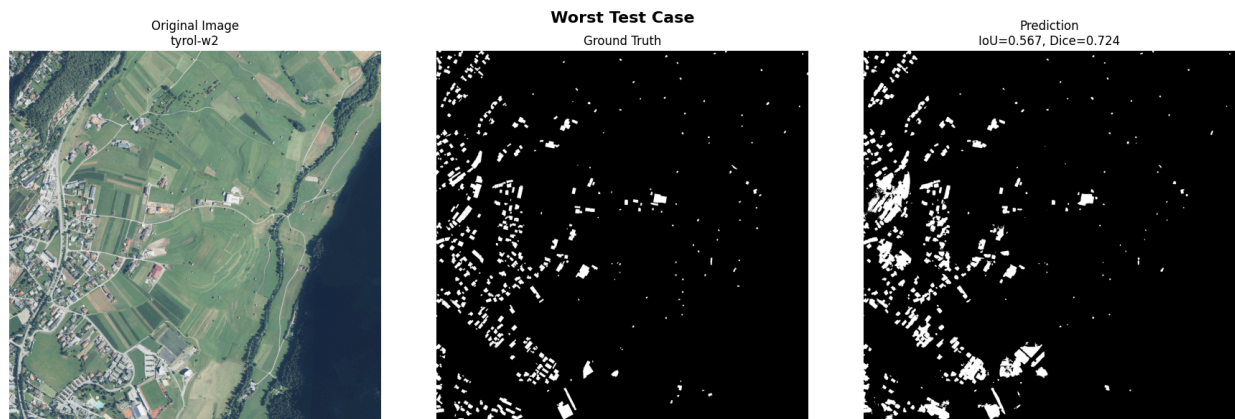
The qualitative examples provide context that the numerical averages cannot show. Figures 10-12 compare the original aerial image, ground-truth mask, and prediction for the best, representative, and lowest-IoU Tyrol cases. The best case (tyrol-w19) reaches  $\text{IoU} = 0.803$  and closely follows the true building pattern. The representative case (tyrol-w32) reaches  $\text{IoU} = 0.704$  and shows the model's typical behavior. The lowest-performing case (tyrol-w2) reaches  $\text{IoU} = 0.567$  and contains more segmentation errors in a sparse rural setting. Figure 13 maps these errors directly, making the false-positive and missed-building regions visible rather than relying only on summary metrics.



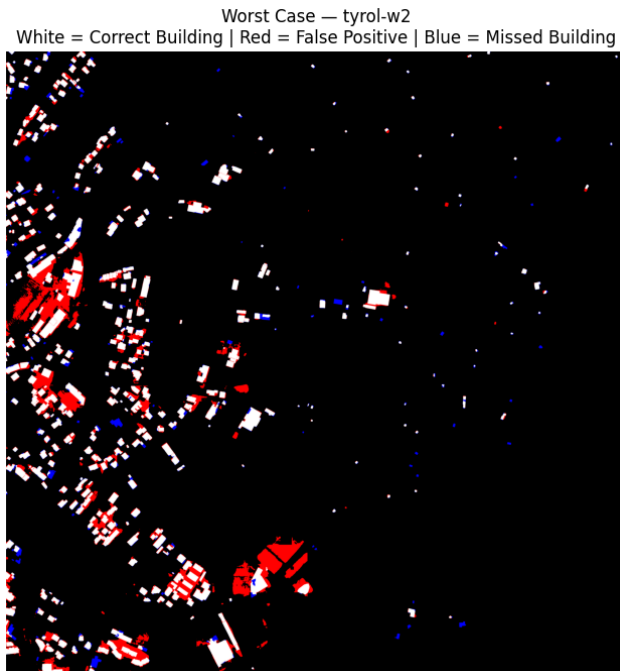
**Figure 10.** Best case — tyrol-w19 ( $\text{IoU}=0.803$ ,  $\text{Dice}=0.891$ ): dense village center, sharply resolved.



**Figure 11.** Representative case — tyrol-w32 ( $\text{IoU}=0.704$ ,  $\text{Dice}=0.826$ ): a typical mid-range tile mixing dense and sparse settlement.



**Figure 12.** Worst case — tyrol-w2 ( $\text{IoU}=0.567$ ,  $\text{Dice}=0.724$ ): sparse rural buildings scattered across open farmland — the hardest regime for this task, since isolated small structures with little regular spatial pattern are both easy to miss and easy to over-predict from nearby texture (roads, field boundaries).



**Figure 13.** Error map for the worst case (tyrol-w2): white = correct, red = false positive, blue = missed building. Errors of both kinds cluster in the same sparse-settlement areas rather than being uniformly distributed — consistent with the precision/recall trade-off seen throughout this test set.

The quantitative and visual results point to the same next step: improving precision and building-edge quality. Recall is already relatively high, so simply making the model more sensitive to buildings is unlikely to address the main weakness. Future work could test a boundary-aware loss, stronger augmentation, post-processing, or more geographically diverse training examples to reduce false positives and sharpen footprint boundaries.

## Part 2 Interpretation and Final Decision

The results show that the model can identify building footprints even in a city that was completely unseen during training. Holding out Tyrol for final testing made the evaluation more realistic and ensured that the model was learning general building features rather than memorizing locations or image patches. The Epoch 13 U-Net + ResNet34 model was selected because it achieved the best validation IoU before signs of overfitting appeared. On the 36 unseen Tyrol images, the model achieved IoU = 0.686 and Dice = 0.814, showing good overall agreement between predicted and actual building areas. The higher recall (0.875) than precision (0.760) gives an important insight into the remaining errors: the model detects most real buildings, but sometimes predicts extra areas as buildings. The visual error maps support this observation, particularly around difficult building boundaries and sparse areas. Overall, the selected model provides a good starting point for building-footprint extraction on unseen locations. Based on the results, the next improvement should focus mainly on reducing false-positive regions and improving building boundary precision, for example through boundary-aware loss or improved post-processing. The table below summarizes the final pipeline and modeling decisions:

| Component    | Choice & justification                                                                                               |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Data         | Full official Inria dataset (180 labeled tiles, 5 cities), run on Google Colab (Tesla T4 GPU)                        |
| Partitioning | Entire city (Tyrol) held out for final test; per-city train/validation split on the rest; zero tile leakage verified |
| Architecture | U-Net + ImageNet-pretrained ResNet34 encoder (24.4M params)                                                          |
| Loss         | BCE-with-logits + Dice (equal weight) — addresses class imbalance (~18% building pixels)                             |

| Component | Choice & justification                                                           |
|-----------|----------------------------------------------------------------------------------|
| Inference | Sliding-window stitching (512px patch / 384px stride) for full 5000×5000px tiles |
| Metrics   | IoU, Dice/F1, Precision, Recall, Accuracy - global pixel-pooled and per-tile     |
| Result    | Global IoU 0.686 / Dice 0.814 on 36 fully held-out Tyrol test tiles              |